

A photograph of a person climbing a rock face, overlaid with various physics diagrams and equations. The diagrams include:

- A vector equation $\vec{\omega} = \frac{d\vec{\theta}}{dt}$ near the climber's head.
- A vector equation $\vec{F} = m\vec{g}$ near the climber's feet.
- A vector equation $\vec{M} = \sum \vec{F} \times \vec{d}$ near the climber's feet.
- A vector equation $[\vec{L}] = N \cdot s$ near the climber's head.
- A vector equation $[\vec{L}] = N$ near the climber's feet.
- A vector equation α near the climber's feet.
- A vector equation $SMO_2\%$ near the climber's head.
- A vector equation $[\vec{d}] = cm$ near the climber's head.

8

WHAT CAN I OPTIMISE IN MY TRAINING SESSIONS?



DEFINITION OF TRAINING GOALS

'If you don't know where you're going, any road will get you there.' This saying highlights the importance of knowing what you're training for. The definition of training goals is the first step to knowing what you should train. Are you preparing for the winter season? The sport climbing season? Is your project a big wall? A boulder problem?

In practice, and except at the most elite levels, the majority of climbers do a bit of all types of climbing (although they usually have a preferred discipline), and they either project individual routes or tick off lots of different routes,

progressing climb by climb. Not planning this progression or selection of climbs can mean that if at the end of the season you decide to try a certain route, you won't have done the right type of training for a successful ascent.

How can I set my training goals? I know that this can be quite hard to do, and we don't always know in advance what we might want to climb. The following example shows one way to structure your goals to help assess, on various levels, where you're at and what you need to work on, so you can design your goals for the season:

- *End goal*: what you want to climb by the end of the season. For example: *Rabadá-Navarro*, a classic multi-pitch in the Picos de Europa in Spain: 750 metres, 21 pitches, F6c+. A very long limestone route with the hardest pitches at the beginning and a well-known traverse halfway up. You'd train very differently for this route if the hardest pitches were right at the top or if it were a granite corner or slab.
- *General goal*: for success on your chosen route, you need to plan your training so that you're ready and able to climb it by the end of the season. If your route has two F6c and F6c+ pitches and the rest are between F6a+ and F4, you should aim to be redpointing at least F6c+/7a, onsighting at least F6b+/6c and F6a should feel like continuity training.
- *Intermediate goals*: these are routes or exercises that you'd do at set points in the season to train for specific features of your chosen route. For example, what characterises the difficulty of the crux pitches on your route? Is it a certain type of move or hold (crimps, friction-dependent moves, two-finger pockets, sustained climbing with no rests)? Are there any other parts that you might struggle with (do you need to work on traversing, for example)?
- *Specific goals*: these are a further breakdown of our intermediate goals. They could include things like not cutting loose on overhangs, getting stronger on two-finger pockets, foot swaps, and so on.

Now you know where you want to be, the next question is where are you starting from? Your starting point, together with how much time you have, will determine how you go about achieving your goals.

To properly plan your training, you need to objectively measure what it is that you need to improve. And as we all know: if you can't measure it, you can't improve it. As such, you need to ask yourself three questions: What do I want to measure? How do I measure it (we need to know the reliability of the methods we're going to use)? Is it relevant to what I want to achieve? For example, it's no good measuring how fast you can dribble a basketball, regardless of how reliable your methods are, if it has absolutely nothing to do with your goals. Let's look at some examples:

- **Example A**
What do I want to measure? Finger strength.
How do I measure it? Using a dynamometer or a deadhang test.
Is it relevant? Dynamometry is a scientifically proven way to measure the concentric strength of the hand flexor muscles but is this type of contraction really relevant to climbing? Or is isometric strength, measured by doing a deadhang test, more relevant?
- **Example B**
What do I want to measure? Pulling strength.
How do I measure it? By working out how many pull-ups I can do.
Is it relevant? We know that climbers need a basic level of pulling strength, but is there any correlation between the amount of pull-ups I can do and the grade I can climb?

In addition to analysing specific details, we should also look at performance in a broader sense. Let's look at finger strength again: imagine a climber who really struggles with short, crimping overhanging routes. They think their problem is finger strength and start training to fix this. After a while, they can hang a 14-millimetre edge for 57 seconds, which, according to López-Rivera (2014d), corresponds to a high level of finger strength. However, when they get back on the same routes, they keep falling off, just like before. If they've got better at what they thought was holding them back, what's going wrong? This is where we need to look at the bigger picture. Maybe their fingers were strong enough for these routes and the limiting factor was a lack of core strength to keep their feet on the wall. Maybe poor mobility meant they couldn't get into the best positions to use the crimps. Maybe they were getting too pumped too soon and the problem was with short-duration endurance.

Proper analysis of your starting point means you can select the best route to achieving your goals and have the best chance of success.

Now you know where you're starting from, let's get moving!

GENERAL WARM-UP. JOINT MOBILISATION. COORDINATION-BASED CARDIO. INCREASING ROM. MUSCLE ACTIVATION

We all know from gym classes or PE lessons that warming up is very important. But animals don't warm up before going hunting, so why should we warm up before climbing or sport? For better or worse, the human body works differently to other animals. We're hugely affected by body temperature (our metabolism is simply a chain of chemical reactions, which work much better with a bit of heat). For a more scientific explanation, the findings of a meta-analysis performed by Fradkin *et al.* (2010) speak for themselves: a proper warm-up affects 79 per cent of performance-related factors and can improve performance by up to 20 per cent.

Returning to the earlier topic of fascia ([chapter 5](#)): a good warm-up will increase the viscosity of deep fascia, improving the sliding of muscle fibrils and the transmission of information, force and muscle movement. It also helps achieve the right degree of fascia stiffness.

Now here's the question we've all been waiting for: do we really know how to warm up for climbing?

As this book is all about training, we'll mainly focus on warming up for training at an indoor wall. The foot of the crag isn't always the best place to warm up, so just try to recreate as much of your indoor warm-up as you can. If this isn't possible, aim to start climbing or moving with the principles and ideas of your warm-up in mind.

To begin, there are three basic aims of a good warm-up:

- To increase body temperature, proprioceptive sensitivity and central stability.
- To refine movement patterns and mental focus for the session.
- To avoid any muscular, psychological or metabolic fatigue that would interfere with the aims of the session.

A good warm-up can be divided into two parts: a general warm-up and a specific warm-up. A general warm-up prepares the body for movement (to make sure you do this, keep your trainers on when you get to the wall and only switch to climbing shoes when you're ready to climb). A specific warm-up does the same but with climbing-specific motor skills and movement.

A good general warm-up begins with joint mobilisation. You could start at the ankles and work up to the head and arms, or vice versa. The aim is to move your joints in all possible directions, while keeping the muscles gently engaged. This will loosen up the joints, get them into the best possible position, improve mobility and stimulate the flow of synovial fluid.

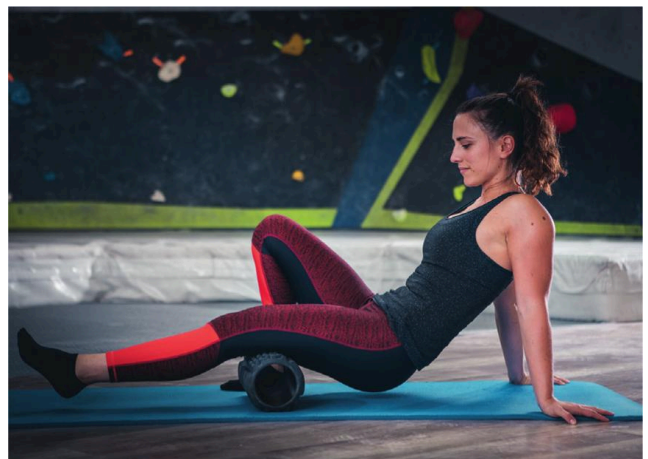
With your joints now prepared for the session, it's time to raise your body temperature in the simplest possible way: with a little bit of cardio. Four to five minutes are enough. Running, high knees, star jumps, exercise bike, skipping, and so on. Keep the intensity at 40 to 50 per cent so you don't get tired. If you can, try to add some coordination work (for example, star jumps raising a different arm with each jump, or high knees circling your right arm on every third raise of your left knee ... use your imagination!). This will help to refine movement patterns, to improve general coordination and to focus your attention on your training.

Now you need to increase the functionality and stability of your joints in all possible positions and angles. We're talking mobility again, just like in [chapter 3](#). There are several different options for improving ROM during your warm-up:

- Stretching*: as we already know, it doesn't make much sense to do any passive or static stretching (which triggers the autogenic inhibition reflex). Instead, you should be doing active or dynamic stretching, holding each stretch for less than eight seconds to avoid activating the autogenic inhibition reflex. The number of reps/contractions/bounces is up to you. Go with what you need, how your body responds and how you're feeling that day. Some days, six to eight reps are enough, and others you might need up to 15.
- Foam rollers, massage sticks, balls ...* : around 8 to 10 passes are enough to inhibit the transmission of nociceptive signals to and from the brain and you'll see an improvement in ROM. Where should you do this? Focus on areas with the least ROM, wherever you feel the most tightness or stiffness. Think about where you might need better ROM, for example, calves, hamstrings, glutes, hip abductors, psoas, between the scapulae. If you have a GDS muscle chain assessment, I'd recommend focusing on your causal and reactive chains (see [images on the next page](#)).
- For areas where your ROM is particularly limited (for example, if you can't properly rotate your shoulder or it hurts when you do, or if your hip abductors need some attention), you could use local vibration combined with isometric contractions held for five seconds each. Five contractions might be enough or you might need more, depending on how you feel and your individual needs. Don't overdo it, though: you're aiming for activation, not fatigue!



FOAM ROLLING CALVES



FOAM ROLLING HAMSTRINGS



FOAM ROLLING GLUTES



FOAM ROLLING BETWEEN SCAPULAE



FOAM ROLLING HIP FLEXORS



FOAM ROLLING ABDUCTORS

Now you can move on to muscle activation. Start with general or multi-joint exercises such as dips, pull-ups, TRX rows, squats and lunges. Then switch to more targeted or single-joint exercises, working the biceps, triceps, and so on, which you can do using resistance bands. Remember to adjust the intensity of these exercises to ensure activation, not fatigue. There are two muscle groups that deserve a special mention here: the hamstrings and the rotator cuff. We often forget to specifically activate the hamstrings, but a nasty shooting pain mid heel hook will soon remind us that they don't like being left out. As for the rotator cuff, you may have seen climbers grab their resistance bands and vigorously work the external rotator muscles 'to prevent injury'. However, warming up and strength exercises for injury prevention are two very different things. When the rotator cuff is fatigued, it's less able to secure the humerus in the glenoid cavity (Chopp *et al.*, 2010), causing joint instability and therefore increasing the risk of injury.

CORE ACTIVATION

Core strength is fundamental in climbing, and the need for core training goes without saying. It's essential for any climber looking for even the most basic level of performance and injury prevention.

Is the warm-up the right time to focus on core? It is, but with the caveat already used in this section: you need to activate, not work or fatigue, the core. You can do this through isolated exercises (such as plank, Pallof press or ab roll-outs) or through climbing exercises that require a little extra core activation (Tic Tac Toe, Feet forwards or Get'em!). But which is better: isolated exercises or climbing exercises? My recommendation is simple: do a few isolated exercises to activate the muscles, followed by some functional exercises that include actual climbing moves where you need to use your core.

At the end of this chapter, you'll find a section on physical conditioning that contains lots of different exercises, including the following core exercises that you could use in your warm-up:

- Plank
- Pallof press
- Ab roll-outs or TRX roll-outs
- Farmer walk
- Cat-cow
- Glute bridge
- Kneeling superman

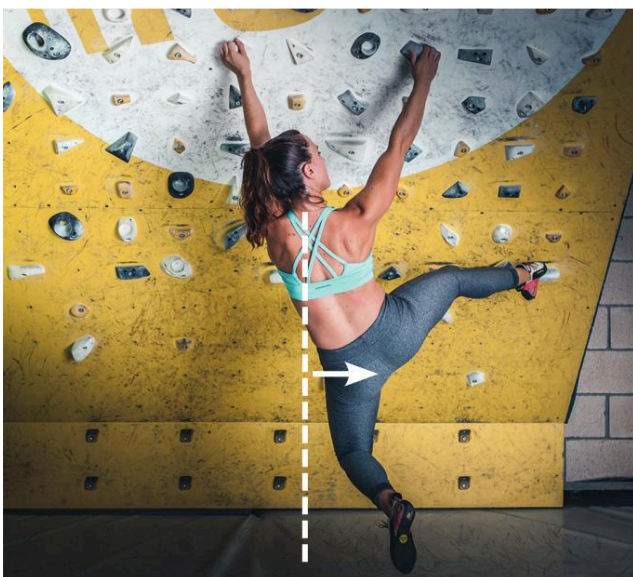


SPECIFIC WALL-BASED EXERCISES

TRAVERSING. INTRODUCING THE SESSION'S TARGET TECHNIQUE

The specific warm-up is what most of us skip to at the start of a session. We get straight on the wall, do some traversing, easy climbing, and so on.

The aim here is to activate climbing-specific patterns of movement, activating your muscles for the movements inherent to climbing. So let's really make the most of this part of the warm-up. What are you planning to train today? Campus boarding? Crimpy problems? Endurance? Tailoring your specific warm-up to the focus of your session is a much more efficient way to train. If you're going to train problems with big moves, campusing or dynos, it doesn't make sense to warm up on small crimps and balancey moves, or to avoid having to pull. It would be much more efficient to do traverses, problems or moves that gradually activate the main pulling muscles and to work hand-eye coordination on increasingly difficult dynamic moves. This targeted activation of the most relevant movement patterns and muscle fibres will help you get the most out of your training session.



TIC TAC TOE

1



TIC TAC TOE

2



TIC TAC TOE

3

WALL-BASED CORE WORK. RESISTANCE BAND EXERCISES AND BODY TENSION

The next step is specific core activation through exercises integrated into climbing movement. Again, remember that the aim is to activate, not strengthen. Matros *et al.* (2013) suggest the following exercises:

- *Tic Tac Toe*: hang from a hold, ideally on an overhanging wall, and tap the furthest possible hold to your right-hand side with your right foot at knee height (remember that you need to move your hips to properly activate the core) and return to centre. Then tap a hold at hip height and return to centre, and finally tap a hold at head height and return to centre. Link this final move into beginning the exercise on your left-hand side with your left foot but this time in reverse order (head height and centre, hip height and centre, knee height and centre).
- *Diagonal*: on a very overhanging wall, or better still a roof, hang for 10 to 15 seconds using just one hand and the opposite foot. The further apart the holds, the harder the exercise.
- *Get'em!*: like Diagonal, you'll need a steep overhang or a roof. Jump up and grab a hold above your head. Swing your feet towards the wall and try to touch (and stay on!) the furthest possible foothold. To increase the intensity, try a running jump to create extra momentum as you jump and swing.



DIAGONAL



HANG AROUND

1



GET'EM!

1



HANG AROUND

2



GET'EM!

2



HANG AROUND

3



TRAVERSING WITH A RESISTANCE BAND AROUND THE HIPS



TRAVERSING WITH A RESISTANCE BAND AROUND THE FOOT

- *Hang around or Cut loose:* on an overhang or a roof, choose a problem or a traverse of medium difficulty. After each move, cut loose with both feet and put them back on again.
- *Feet forwards:* in this exercise, touch the next handhold with your foot before getting it with your hand. Repeat on each move. The steeper the climb, the easier the exercise (see [images on next page](#)).
- *Traversing with a resistance band around the foot or hips:* in pairs, one climber traverses while the other pulls on a resistance band looped around the climber's hips or foot.
- *Body tension boulder problems* (see images on [following pages](#)): on this type of problem, both your hands and feet should be a long way from your centre of gravity, meaning you need a lot of full body tension to not fall off. Another way to recruit full body tension is through an exercise called Freeze:
 - Find a problem with no dynamic moves and 'freeze' in position for two to three seconds after each move. This exercise not only works core strength but also develops lock-off strength, technique and movement

efficiency.



FEET FORWARDS

1



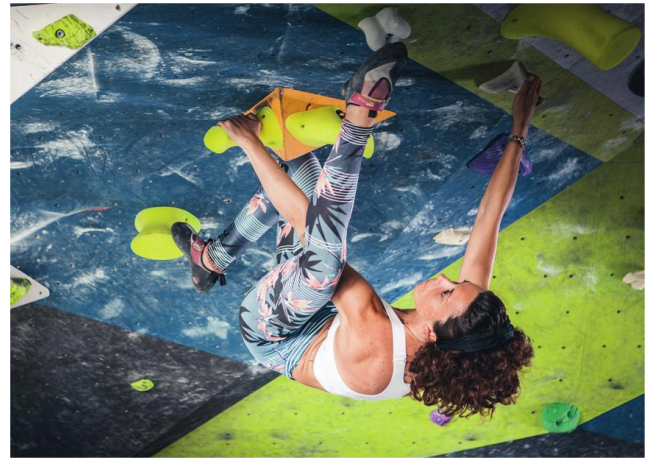
FEET FORWARDS

2



FEET FORWARDS

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FEET FORWARDS

4



FEET FORWARDS

5



BODY TENSION PROBLEMS

1



BODY TENSION PROBLEMS

2

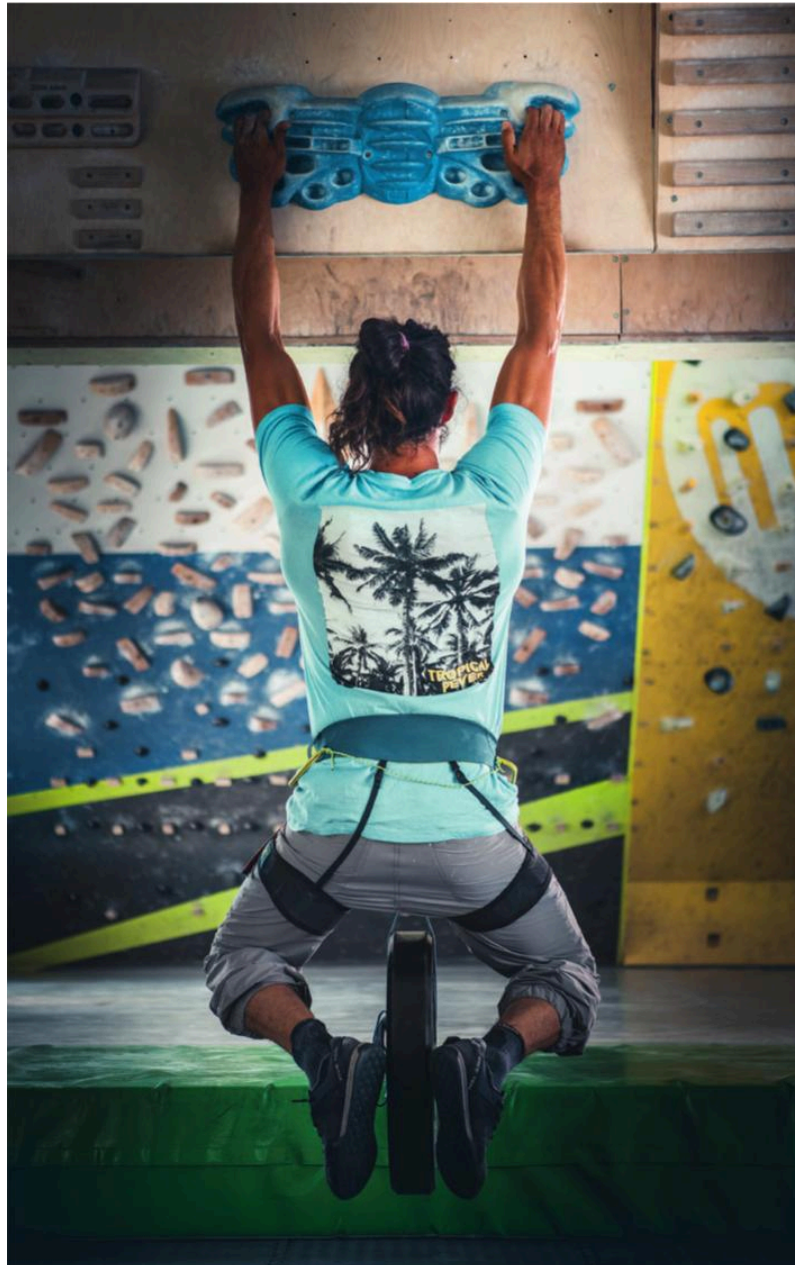
ENHANCING PERFORMANCE: POST-ACTIVATION POTENTIATION

Post-activation potentiation (PAP) can lead to a high degree of neural stimulation and result in better motor neuron recruitment, greater firing rate and improved interaction of actin and myosin muscle fibres (Marchante, 2015). In other words, it not only warms up your muscles, but gives the nervous system that touch of maximum activation to maximise your training performance.

It involves doing exercises specific to your training session (for example, if you're going to train hard crimping problems, you'd use deadhangs for your PAP) at a very high intensity but without causing fatigue. PAP training can be applied to both dynamic exercises (such as weighted pull-ups, jumping pull-ups, jumping dips and box jumps) and isometric exercises (such as max weight hangs on a large edge, minimum edge hangs or weighted lock-offs at different angles).

How does PAP work? The first thing to know is that PAP only works for very advanced climbers as it targets fast-twitch muscle fibres (it won't have any effect if you are a beginner). The following points are essential for effective PAP training:

- *Intensity*: you have to be working at maximum intensity – higher than 85 per cent of your 1RM, your max weight hang on a large edge or the smallest edge you can hang.
- *Number of sets*: depending on ability, you could do between one and five sets, resting for around three minutes between each set.
- *Repetitions and time under tension*: anything over five repetitions, or isometric contractions for longer than five seconds are likely to cause fatigue. For optimal results, do no more than three to four repetitions and hold isometric contractions for no longer than three seconds.



After PAP training, you'll need to rest for 7 to 12 minutes before continuing with your session. However, don't just stand still during this time as you might cool down and lose all the work you've done warming up. Make the most of this time to keep moving on the wall or to focus on your weaknesses.

A good warm-up will take up a lot of your training session. So, you might be tempted to spend more time training and less time warming up. However, following the advice and philosophy of this book ('don't train more: train better'), you'll see that an effective training session can be much shorter than you might think. And as the saying goes: it's best to keep it short and sweet.

CLIMBING-SPECIFIC TRAINING NEEDS

Before revealing the 'recipe for success', which many readers will be looking for from this book, I'd like to define some useful terminology.

Methods are the processes that define an exercise, the means that are used and the applicable training load. For example, we could use the velocity-based training method when doing pull-ups to train pulling strength.

Means are the equipment or apparatus used to implement a training method. For example, the campus board is a means (not a method) of training power. And there are several means of training endurance: an outdoor route, an indoor route or a MoonBoard.

Training volume refers to the total amount of training activity, measured in time, distance covered, total number of moves, and so on.

Intensity is the qualitative measure of the training load: the more work per unit of time (for example, metres climbed in one minute, or weight lifted in one rep), the higher the intensity. It's determined not only by mechanical effort, but by energy used in the nervous system and psychological load.

Lastly, *training density* is the amount of work time versus the amount of recovery time: for example, one minute of work × two minutes of rest; two days of training × one day of rest ...

MAXIMUM STRENGTH TRAINING

As mentioned in previous chapters, maximum strength is the maximum amount of force that can be applied in a specific action or exercise. It's vitally important in any sport and especially in climbing, where there are three areas where performance is dependent on maximum strength:

- *Maximum grip strength*: as in any other sport, climbers need to do isolated training exercises, where all elements can be controlled without the multiple/additional stimuli of actual climbing. This allows us to apply as much force as possible and is why deadhangs are used to improve maximum grip strength. However, for the strength gained in these exercises to be useful, it needs to be transferred to climbing, and this is where bouldering comes in.
- *Maximum pulling strength*: the maximum amount we can pull (and hold) with our arms. A good level of pulling strength helps with very physical moves, lock-offs, overhangs, and so on. We can train pulling strength by doing weighted pull-ups, lock-offs and heel hooks.
- *Maximum pushing strength*: often in climbing, despite being in the right position, we're not able to 'use our feet' to weight them properly and push with our legs to do the move. This also applies to pushing strength in our arms on mantels where we don't have the strength to push our way through the move. This is why we need to train pushing strength by doing deadlifts, squats, pistol squats (one-legged squats), bench presses and dips.

At the risk of sounding like a broken record, maximum strength training is so important in climbing because:

- *It reduces the risk of injury*: as we saw in [chapter 2](#), strength training is the most efficient type of training for injury prevention, and training at over 80 per cent of 1RM results in greater strength gains. It also makes a lot of sense from a structural perspective: if you can do a weighted hang on a 10-millimetre edge, pulling on a 16-millimetre edge is going to be much easier on your tendons than if you can barely hang a 14-millimetre edge. Once again, maximum strength increases submaximal strength; think back to our analogy of the 70-horsepower car and the 500-horsepower Ferrari. Greater levels of maximum strength mean less stress on your tendons.
- *It improves performance*: the stronger you are, the harder you can climb: You can use smaller handholds and footholds, apply more force through these holds, pull harder, do bigger moves, keep your feet on more often and get them back on much easier ... What's more, the further away you are on each move from maximum effort, the easier each move will feel, so your endurance will also improve.

Is there a risk of injury when training maximum strength? The answer is YES. Max strength training exposes muscles and tendons to a huge amount of mechanical strain. The injuries that could occur include:

- Torn muscle belly: the least likely to happen.
- Tear at the muscle/tendon junction or tendon/bone junction.

But this is why you should work progressively. You should never dive straight into weighted deadhangs or squats at 90 per cent of your 1RM. Training is all about process, adaptation and progression. In any case, the following guidelines can significantly reduce the risk of injury in maximum strength training.

For general strength exercises (Benito, 2008):

- a) A minimum of 18 months (consecutive, not spread over 10 years) of strength training is recommended before training at between your 1RM and your 3RM (three-repetition maximum: the maximum amount with which you can perform three repetitions).
- b) With less experience, it's better to do higher reps and lower weights (three to five repetitions).
- c) Always do multi-joint exercises: squats, pull-ups, deadlifts, bench press, and so on. Attempting a 1RM bicep curl is like begging for a ruptured tendon.
- d) Always use proper equipment: make sure there's a mechanism or a spotter in place in case you can't complete a lift. Power racks or Smith machines are perfect for training with heavy weights.

For climbing-specific exercises (deadhangs):

- a) A minimum of two years of uninterrupted training (two to three sessions a week) is recommended (López-Rivera, 2018a).
- b) With less experience, strength should be built by climbing (López-Rivera, 2018a) or by hanging on a pull-up bar or on a minimum 24-millimetre edge/three-pad hold (Hörst, 2006/2004).

MAXIMUM GRIP STRENGTH. DEADHANGS. EVIDENCE AND PROTOCOLS. ADAPTATION FOR DIFFERENT LEVELS OF TRAINING

Edge size	Time (seconds)	Level
14mm	0–30	Low
	30–50	Medium
	50+	High



We've finally reached the part you've all been waiting for: how to train, what to do ... the magic formula. The first thing to consider is which is the most efficient: weighted deadhangs on a large edge, or deadhangs on the smallest possible edge with no added weight. Thanks to high-quality research, notably Eva López-Rivera's PhD thesis (2014d), we can answer this with proven scientific evidence.

Participants in the study conducted by López-Rivera did two different tests:

1. *Strength test*: hang on a 15-millimetre edge for five seconds with as much added weight as possible.
2. *Endurance test*: hang on an 11-millimetre edge for as long as possible with no added weight.

The participants were then divided into two groups:

- *Group A*: this group started with four weeks of 10-second max-weight hangs on an 18-millimetre edge. They rested for two weeks and then did four weeks of 10-second minimum-edge hangs (hanging on the smallest possible edge with no added weight).
- *Group B*: this group did the same but in reverse order. They started with four weeks of 10-second minimum-edge hangs. They rested for two weeks and then did four weeks of 10-second max-weight hangs on an 18-millimetre edge.

The study found that:

1. Compared to initial levels, the greatest changes in strength and endurance occurred after four weeks of training in both groups.
2. There was a significant positive correlation between improved grip strength and grip endurance, adjusting for body weight.
3. The most efficient way to improve grip strength and endurance is to first train max-weight hangs on a large edge and then train minimum-edge hangs.

López-Rivera (2018b) proposes the following protocol for max-weight hangs on a large edge:

- 3–5 max-weight hangs × 8–20mm × 5–15 seconds (1–5): 3–5 minutes

In other words, do three to five max-weight deadhangs on an edge between 8 and 20 millimetres deep, hanging for five to fifteen seconds, at an effort level (EL) of one to five seconds, resting between hangs for three to five minutes. This formula gives a range of combinations for different abilities, allowing for controlled progression.

For minimum edge hangs, López-Rivera (2018b) proposes:

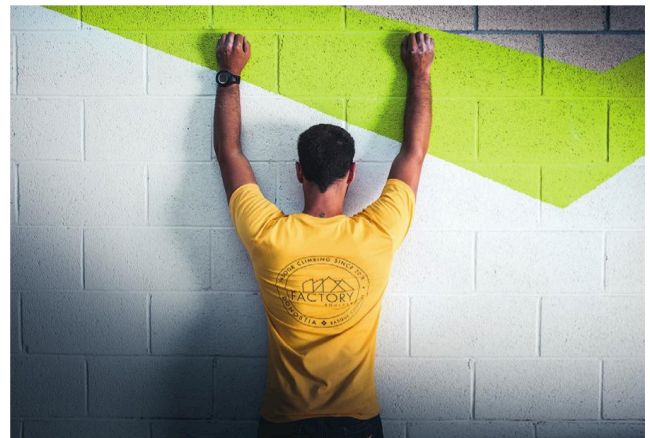
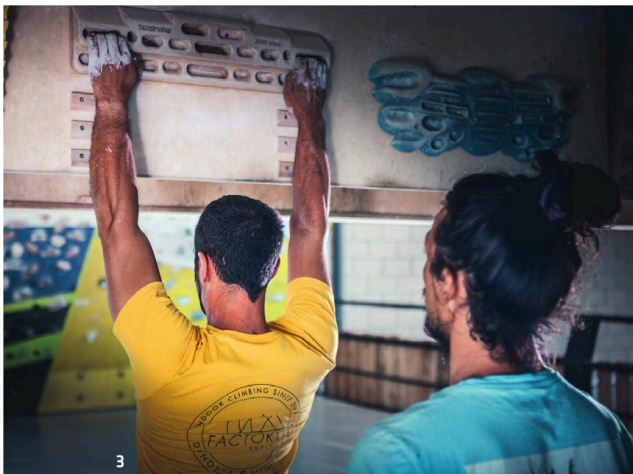
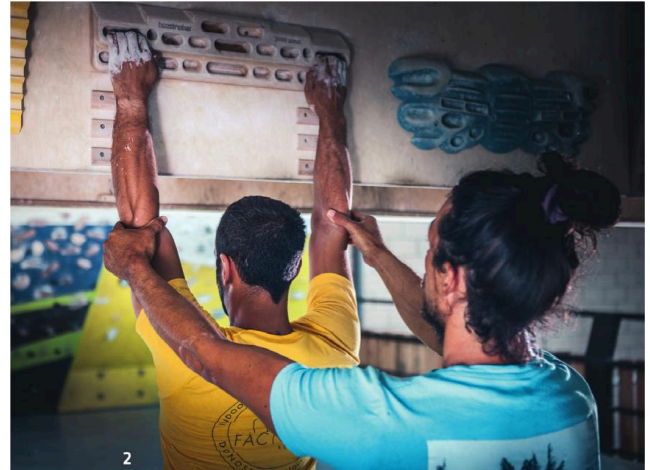
- 3–5 hangs on minimum edge × 5–15 seconds (1–5): 3–5 minutes

In other words, do three to five deadhangs on the smallest edge that you can hold for five to fifteen seconds, at an EL of one to five seconds, resting between hangs for three to five minutes.

The effort level (EL) in the López-Rivera protocols represents the difference between the number of seconds you hang for and the maximum number of seconds you *could* hang for. That is, it's your margin before failure or the number of seconds held in reserve. As a way of measuring intensity based on personal perception, it's perfect for adapting intensity to any individual or level of ability. Other authors express effort level as: number of reps done (maximum number of reps possible). For example: 5(10). This is also referred to as level of intensity or perceived effort.

Another leading voice in training for climbing, Eric Hörst, proposes the following protocols (2016):

- Training on a large edge (14–20mm) with maximum added weight:
Two sets (where one set = five hangs × 10 seconds on an edge you can barely hold for 13 seconds at maximum effort x three minutes of rest between hangs) with five minutes of rest between sets.
- Training on a minimum edge with no added weight:
Two sets (where one set = five hangs × 12 seconds on an edge you can barely hold for 15 seconds at maximum effort x three minutes of rest between hangs) with five minutes of rest between sets.



CORRECT ALIGNMENT FOR DEADHANGS.

The following climbers **should not do max weight or minimum edge protocols** and will need to use other training methods to build maximum strength without injuring themselves (López-Rivera, 2011):

- Climbers with less than two years' experience of systematic training.
- Climbers under 16 years old.
- Climbers with a medium level of strength: able to hang with straight arms for over 15 seconds on a 24-millimetre edge, under 35 seconds on a 20-millimetre edge and under 10 seconds on a 10-millimetre edge.

Hörst (2006/2004) suggests the following alternatives for these climbers:

- Hanging on a pull-up bar until they can hang for one to two minutes.
- Deadhangs on a 24-millimetre edge or three-pad hold on a fingerboard (i.e. a good hold).

Now let's take a look at correct technique for deadhangs (Smith and Blumenthal, 2016):

1. Always ensure good scapular retraction (shoulder blades back and down away from the ears, neck long).
2. Arms should be fully extended, elbows soft. Elbow crease should be facing towards the ears.
3. Head should be in line with the rest of the body, not straining forwards.
4. Use an open-hand grip position (recommended) or half crimp, but never full crimp (thumb over fingers).

5. Don't overextend your wrist. The metacarpal bones (knuckles) should be in line with the shoulder, with no rotation through the wrist.

Are assisted hangs a good idea? The answer, like practically everything to do with training, is, 'it depends':

- If you need to take weight off because the edge that you're using is too small, then obviously the answer is NO. If you aren't strong enough to hang the edge, this will cause excessive strain on your tendons and ligaments with too great a risk of injury.
- If you're taking weight off because you're doing one-finger or two-finger hangs and you want to reduce the strain on your tendons, ligaments, etc., then the answer could be yes. However, you need to think critically and be sure that you're not training beyond your limits. It may well be better to build maximum strength in these fingers by climbing on rock or plastic.
- One good reason for taking weight off is to progress to one-arm hangs. With one-arm hangs, aside from doubling the load on the active hand, the shoulder joint is far more vulnerable and needs protecting from both vertical forces (through scapular retraction) and rotational forces (using the pectoral muscles, rotator cuff and the muscles surrounding the elbow joint). Holding on to a rope with the assisting hand means you can progressively increase the load on the active hand, while also preventing rotation through the shoulder joint. A second option is to use a resistance band, which provides less assistance. A third option is to use a pulley system, which gives selective and precise control over the amount of weight you take off. However, this system won't stop the shoulder from rotating. As a final recommendation, even if you can hang a 10-millimetre edge, you should start doing one-arm hangs on a three-pad hold. This will progressively build the strength needed to stabilise the shoulder joint while not overloading the elbow joint.

MAXIMUM PULLING STRENGTH. PULL-UPS. VELOCITY-BASED STRENGTH TRAINING. MAXIMUM INTENSITY METHODS AND CALCULATING YOUR RM

Pull-ups are the climbing exercise par excellence. Although the ability to do endless pull-ups isn't synonymous with climbing really hard, it's undeniable that you're going to need to pull with your arms on certain moves. And, as we've said many times in this book, the greater your maximum strength, the easier it is to make a move – in this case, pulling yourself up with your arms.

Research carried out by Dr González-Badillo and his team has led to the development of velocity-based training (VBT). This research found a 0.96 to 0.99 correlation (r^2) between the mean propulsive velocity (MPV) at which someone can lift a certain weight and the relative intensity that this entails. That is, a nearly 100 per cent reliable method of using velocity to measure intensity. But what does this mean? It means that if we measure the speed at which we can lift a certain weight (in this case, our body weight, with or without added weight), we can work out the exact intensity at which we're working.

Sounds good so far. In relation to building strength, this type of training is the ultimate expression of my training philosophy, 'don't train more: train better'. To build maximum strength through myofibrillar hypertrophy, stimulating neural pathways to maximise recruitment of muscle fibres, we know you need to be working at maximum intensity (80 to 85 per cent of 1RM). We can use a range of devices (T-Force, VITRUVÉ, Beast Sensor, Push Band, Velowin) to measure MPV for pull-ups. And we can use the following table to calculate intensity based on our MPV:

Intensity %	MPV
50	1.11m/s
55	1.02m/s
60	0.93m/s
65	0.84m/s
70	0.75m/s
75	0.66m/s

Muñoz-López *et al.* (2017)

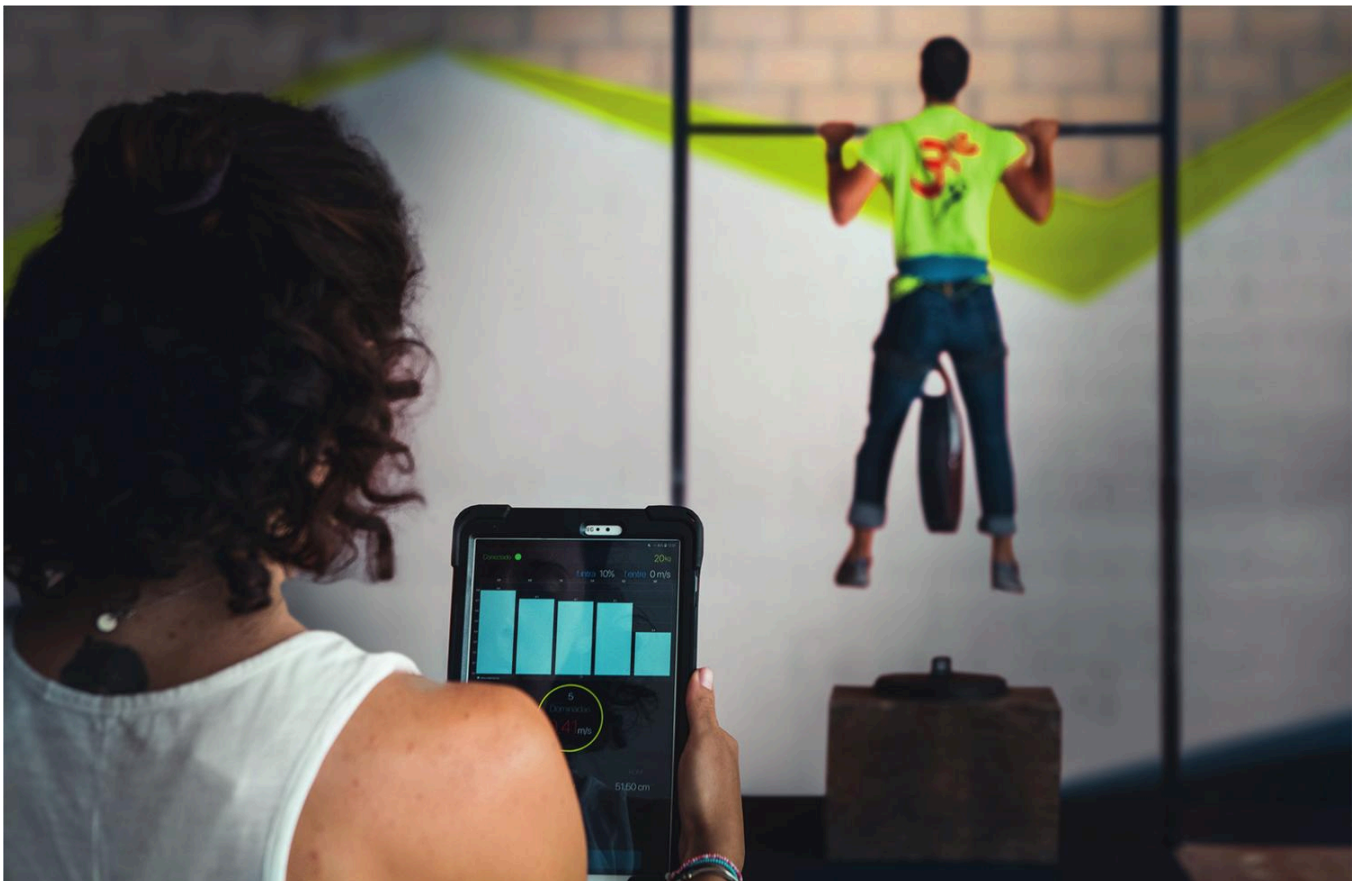
Intensity %	MPV
80	0.57m/s
85	0.48m/s
90	0.39m/s
95	0.30m/s
100	0.21m/s

Begin by measuring how fast you can do the upward pull of a pull-up (concentric phase) and add extra weight until your MPV coincides with the intensity at which you want to work. Remember that you need to be pulling as fast as you possibly can.

Once you know the amount of weight to add, you can start training. Another key benefit of VBT is that it allows you to objectively quantify and reduce fatigue. But why is it so important to reduce fatigue? The aim of training is to train at maximum intensity for the maximum amount of time, but the more fatigued we get, the harder it is to maintain maximum intensity. If you reach a level of fatigue in one session that stops you from giving 100 per cent in the next, you're missing an opportunity to improve. In other words, you need to train at maximum intensity for just the right amount of time so you don't get too tired and can give it your all in the next session. If your MPV dips (i.e. your pull-up is slower) after a certain number of reps or sets, it means that you're finding it harder than you should to lift the same load and that, although you're working at maximum effort, your neuromuscular system just isn't able to produce the same amount of strength. As discussed in the [section on training to failure](#), a 40 per cent reduction in MPV compared to your first rep brings you dangerously close to failure, leading to excessive fatigue and consequences that are simply not compatible with efficient training. It's all about training as little as possible for the greatest possible results. As soon as you're not training at maximum intensity you should stop, as you won't get the results that you're working so hard to achieve. In strength training, there should be a loss of no more than 15 to 20 per cent compared to your first rep. This coincides with approximately half the maximum number of reps you could do with the same load, for example an effort level of 2(4). Therefore, if your MPV at the start of a set is 15 to 20 per cent lower than your initial MPV, it's time to stop doing pull-ups and move on to something else.

What are the actual benefits of VBT compared to more 'traditional' methods? VBT provides a precise method for adjusting the intensity of training, which is a difficult thing to get right for two reasons:

- On the one hand, the external load (the weight lifted over a set number of reps) needs to increase over the course of your mesocycle. For example, you measure your RM on day one and you can do one rep with 40 kilograms of added weight. Then you train for four weeks at 90 per cent of your RM, which is 36 kilograms of added weight. If the training is effective and you get stronger, your RM should have improved by the fourth week. As such, 36 kilograms will correspond to a lower intensity than before, meaning you'll no longer be working at 90 per cent as planned. So, what's the solution? Calculate your RM each week and adjust the added weight accordingly? However, your RM can vary a lot from one day to the next, so a weekly calculation isn't that helpful. What about measuring RM on a daily basis? This would create too much fatigue and detract from your actual training, as well as creating an unnecessary risk of injury.
- On the other hand, and yet closely related to the previous point, your internal load also varies from day to day: you don't always get the same amount of sleep, your eating habits change, stress at work can affect you differently at different times, your motivation isn't always the same, and so on. All these variables can have a huge effect on your ability to generate force and will therefore also affect RM.



With velocity-based training, you can measure the speed of your pull-up in real time, meaning you can calculate the exact intensity you're working at to lift a specific load. You can also adjust the external load depending on your internal load, meaning you can prevent any undue fatigue. What's more, these calculations are based on objective data and offer near 100 per cent reliability. But is this going a little overboard?

And, of course, what if you don't have a VBT device? Unfortunately, these devices aren't cheap (varying between £250 and £4,000) but you could still use the maximum strength methods developed by González-Badillo and Gorostiaga-Ayestarán (2002) before the VBT system. In any case, I would recommend using these VBT principles as a guide:

- Maximum intensity method I (**for advanced climbers**)
 - 4–6 sets × 1–2 reps × 95–100% 1RM × 3–5 minutes of rest between sets
- Maximum intensity method II (**for novice climbers, focus on avoiding muscle failure**)
 - 4–5 sets × 3–5 reps × 85–90% 1RM × 3–5 minutes of rest between sets

To accurately calculate your RM without actually doing a RM, you could use the Brzycki formula. This formula is 99 per cent accurate with a weight that can be lifted 7 to 10 times (Nascimento *et al.*, 2007).

- $1RM = \text{weight lifted} / (1.0278 - (0.0278 \times \text{number of reps}))$

For example, if you weigh 72 kilograms and can do eight full pull-ups (chin above the bar, arms fully extended and scapular retraction maintained) with 18 kilograms of added weight and fail on the ninth pull-up, your 1RM calculation would be:

$$(72 + 18) / (1.0278 - (0.0278 \times 8)) = 111.75 \text{ kilograms}$$

And so to work at your 1RM, you would need to add 39.75 kilograms (111.75 – 72 kilograms).*

What if you can't do a pull-up? It's fairly common to see novice climbers using resistance bands to do assisted pull-ups, but this isn't such a good idea for two reasons:

1. The band gives the most assistance when it's most stretched (at the start of the movement) and takes gradually less weight as it returns to its original length (during the upward pull). Therefore, it could give more help than you actually need at the start of the pull-up and not enough in the middle or at the end.
2. Doing lots of reps with lighter or submaximal weights is not the best way to gain strength.

The eccentric method (just lowering down from the bar) is the most efficient way to work towards a full pull-up. Eccentric contractions generate more force than concentric ones, and more force is exactly what is needed. Standing on a step, bench or chair, position your chin over the bar as if you've just done a pull-up and lift your feet off. Now try as hard as you can to stay in this position. Keep fighting to stay up, even as you start lowering down, as you need to keep the brain signalling to the muscles to pull up. González-Badillo and Gorostiaga-Ayestarán (2002) suggest the following protocol:

- 4–5 sets × 1–6 reps × 100–140% 1RM × 3–5 minutes of rest between sets

As for intensity, if you can't do a pull-up, it suggests that your own body weight is over 100 per cent of your RM as it's a weight that you're not able to lift. If you can comfortably do six reps of eccentric pull-ups but you still can't do the upward pull without assistance, instead of resorting to resistance bands, continue with the eccentric method but start adding weight. How much weight? Add as much as possible providing you can still control the movement, aren't training to failure or lowering down too fast (it should take three to nine seconds to lower down properly).

At the opposite end of the spectrum, very strong climbers may want to progress to one-arm pull-ups. Del Castillo (2019) recommends being able to do 20 pull-ups before starting to train for one-arm pull-ups. Alternatively, PowerExplosive (2013) suggests being able to do between one and three weighted pull-ups with 65 per cent of your body weight. PowerExplosive also warns that pull-ups are very demanding on your biceps so you should focus on strengthening your biceps if they aren't as strong as your lats, forearms, and so on, to reduce the risk of injury. According to Del Castillo (2019), you should start with some form of static support (such as a towel) for your free arm, progressing to elastic support and finally to holding the wrist and then the shoulder of the pulling arm. You can also use the eccentric method, although to begin with you might need someone to support and stabilise you.

You may be wondering what the best grip position is for pull-ups (narrow, wide, neutral, pronated, supinated, and so on). The answer is simple: the aim is to improve maximum strength, so choose the grip where you feel you can generate the most force. That said, a study on scapular biomechanics by Prinold and Bull (2016) found that supinated pull-ups (chin-ups) and wide grip pull-ups can compress the subacromial space, leading to greater risk of subacromial impingement (pinching the supraspinatus tendon). As such, if you have any problems with your shoulders, it's best to go for a narrower grip.

As for proper technique for pull-ups, optimal range of movement and other considerations such as transfer to climbing, the section on physical conditioning '[Strength training to maximise performance. Methods and techniques for upper body, core and lower body exercises](#)' gives a detailed description that should answer all your questions.



MAXIMUM ISOMETRIC STRENGTH. LOCK-OFFS

Unfortunately, the VBT method can't be used to train lock-off strength, as lock-offs are an isometric exercise and so there's no velocity to measure.

It's important to stress that isometric training only increases strength at the exact angle you're training at. For example, if you train lock-offs at 90 degrees, your lock-off strength will only improve at 90 degrees, not at 89 or 91 degrees. As such, you need to include as many different angles and positions as possible and choose positions that are most transferable to climbing. (In my opinion, a two-handed supinated lock-off with your chin over the bar isn't really that useful as you'd be hard pressed to find that position in climbing.)

González-Badillo and Gorostiaga-Ayestarán (2002) also suggest two methods for training maximum isometric strength:

- Isometric RM method:
5 sets × 2 reps (contraction – rest – contraction) × 3–6 seconds per contraction × load 100 per cent 1RM × 3 minutes of rest between sets
- Isometric fatigue method (load closer to 60 per cent 1RM for beginners and closer to 90 per cent for more advanced climbers):
3–5 sets × 1 rep × 12–20 seconds × load 60–90 per cent 1RM × 3 minutes of rest between sets

Interestingly, a study by Schaefer and Bittmann (2017) compared how long subjects could resist an external force to how long they could push against the same force (both isometric actions), working at 80 per cent of their MVC. The subjects could push against the external force for over twice as long as they could resist. Take note!

RECOVERY TIME DURING AND BETWEEN SESSIONS

In terms of recovery, whether between reps, sets or training sessions, there are two types of fatigue to bear in mind: metabolic fatigue and central nervous system fatigue.

Metabolic fatigue reduces creatine phosphate (PCr) stores by 60 to 70 per cent. As we're doing so few reps per set, we're barely using anaerobic or aerobic glycolysis to produce energy, so we'll not look at the depletion of glucose stores. There's an 84 per cent recovery of phosphagen after two minutes, 89 per cent after four minutes and 97 per cent after eight minutes (Billat, 2002). This is why it's so important to rest for three to five minutes between sets in any type of maximum strength training (hangs, pull-ups, lock-offs, and so on), even if you don't feel like you've done that much or you don't feel that tired.

Central nervous system fatigue affects the function of neurotransmitters (mainly acetylcholine), which play a key role in transporting electrical energy through the nervous system to the motor neurons and on to the muscles, where it's converted into mechanical energy. That is, they help transform electrical signals into muscle contractions (Wilmore and Costill, 2007). After maximum strength training, which causes central nervous system fatigue, you'll need between 24 and 48 hours' rest to make sure you can train efficiently in your next session (Zatsiorsky, 1995).

With deadhangs, López-Rivera (2018c) suggests two sessions per week, with 48 to 72 hours between each session for the best results.